

**Prerequisite:** Real Analysis

**Category:** New

**Syllabus:- Module-1: Topology**

- A Topological spaces:- topological space, base, subbase, closed set and limit point, Hausdorff space. [5 lectures]
- B Construction of new topological spaces:- product topology on  $X \times Y$ , subspace topology, metric topology. [3 lectures]
- C Continuous functions:- continuous functions and homeomorphisms. [3 lectures]
- D Connectedness:- connected space, path connected space, connected subspaces of the Real Line. [5 lectures]
- E Compactness:- compact space, Lebesgue number lemma, compact subspaces of the Real Line. [4 lectures]

**Syllabus: Module-2: Complex Analysis**

Complex functions, Continuity, Differentiation, Analytic functions, Cauchy-Riemann equations, [6 L]

Contour integrals, Cauchy-Goursat Theorem, Cauchy Integral formula, Morera's theorem, Liouville's theorem, Fundamental theorem of Algebra and its applications, [8 L]

Isolated singularities, Taylor and Laurent series, Cauchy residue theorem, Evaluation of real integrals [6 L]

**References: References (Topology):-**

- A Topology, Prentice Hall, Inc, 2000, Second edition by James R. Munkers.
- B Introduction to Topology and Modern Analysis, 1983, George F. Simmons, (Robert E. Krieger Publishing company, Inc)
- C A taste of Topology, Volker Runde by S. Axler and K. A. Ribet, Univeritext, Springer, 2000.
- D Schaum's Outline of Theory and problems of General Topology, Seymour Lipschutz, McGraw-Hill Book Company, 1965.
- E Introduction to Topology (Pure and Applied), Colin Adams and Robert Franzosa, Pearson.

**References for (Complex Analysis)** 1. J.B. Conway, *Functions of one complex variable*, Springer

2. L-S. Hahn and B. Epstein, *Classical Complex Analysis*, Jones and Bartlett

3. E.M. Stein and R. Shakarchi, *Complex Analysis*, Princeton University Press

4. T. Gamelin, *Complex Analysis*, Springer

5. James Ward Brown and Ruel Vance Churchill: *Complex Variables and Applications*, McGraw Hill

**Teachers: Prof. B. Bhowmik, Prof. P. Chakraborty**